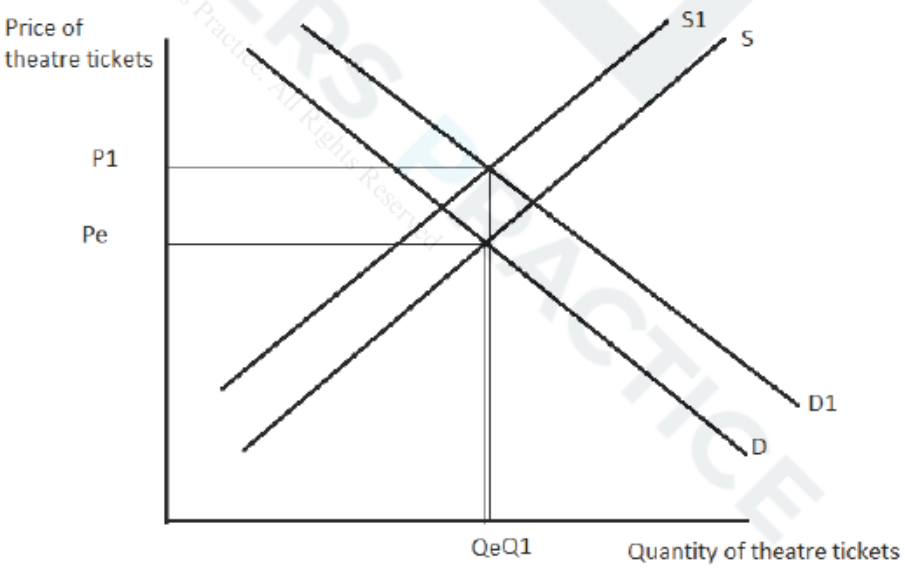
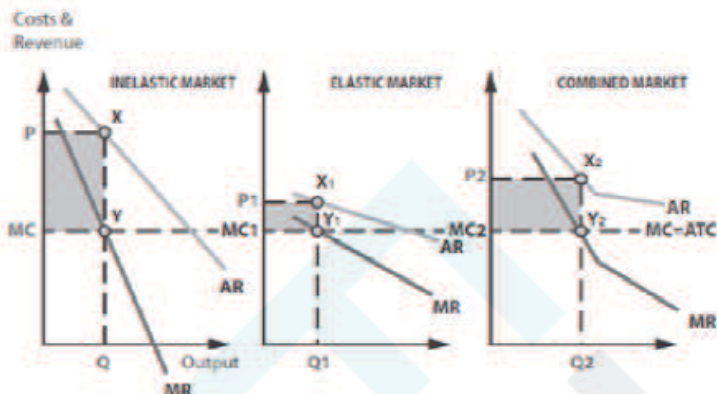


Mark Scheme

Q1.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 1, Analysis 3 Knowledge/understanding: 1 mark for: - accurate supply and demand with original equilibrium. (1) Application: 1 mark for identifying either: - record numbers of people wanting to buy tickets for shows (1) - costs of running the theatres has risen to pay for restoration and absorb the cost of booking fees. (1) Analysis: 1 mark for each of the following: - supply shifted leftwards (1) - demand shifted rightwards (1) - accurate new equilibrium showing higher price. (1) For example: 	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Analysis 2, Application 2, Evaluation 2 Knowledge/understanding: 2 marks for identification of two reasons (1+1). Analysis: 2 marks for linked explanation of these reasons (1+1). Application: 2 marks for reference to the data (1+1), e.g. • to enter seven overseas markets (1) – including Bulgaria, Czech Republic, Romania, Hungary, Israel, Poland, Romania and Slovakia/ operates almost 100 multiplexes (1) which increases the quantity of potential consumers (1) • to establish a European multiplex powerhouse (1) – CCI operates almost 100 multiplexes (1) which increases the quantity of potential consumers at each location (1) • Cineworld was ordered by competition regulators to offload three cinemas following its takeover of the Picturehouse chain (1). Difficulty of finding new growth opportunities in the company's home market (1) so moving abroad enables expansion (1) • industry data by Rentrak showing that UK and Ireland box office takings in 2013 fell by 1% to £1.17bn (1) falling demand in UK (1) and markets in Europe may be growing (1) • rivals have also grown overseas, with Odeon UCI and Vue Entertainment (1) and this suggests is a rational decision for the business to make (1) as their competitors are likely to have moved to maximise profits. (1) Evaluation: 2 marks for two evaluative comments, e.g. • however, some of the countries they are moving to are poorer than the UK (1) • even in these countries they will be scrutinised by EU regulation (1) OR 2 marks for identification and linked development e.g. • magnitude of fall in cinema sales – fall 1% (1) could just be a bad year for films (1) • unlikely to want to copy them (1) it is more the impact on their competitors' profitability they are concerned about. (1)	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 - Define price discrimination. - Application- same screening but different prices – examples from data. - Third degree price discrimination diagram.  (Source: www.economicsonline.co.uk) - If they do not discriminate and $MR=MC$ profit is $P_2MC_2Y_2X_2$. - Adults – demand more inelastic so willing to pay more and respond less as price rises. - Children – demand more responsive to change in price – elastic. - If firms splits up the market and charges different prices to each group the area of profit $PMC_1Y_1X_1$ and $P_1MC_1Y_1X_1$ will be greater than if do not discriminate. - Profits will rise if they price discriminate depending on different groups elasticities.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • Cost to administer and to ensure people buy only appropriate tickets is expensive. • Assumes they know shapes of demand curves for each group – which they may not. • Even within sub groups very different markets with different sensitivities to price – so could increase profits further with other discrimination. • Ethics/fairness of price discrimination. • Magnitude of price variations – £6.80-9.60. • Depends on proportion of earnings from tickets – other earning (popcorn etc) may be more significant. • Cutting costs may have more of an impact on increasing profits than changing price and revenue.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 2, Application 2, Analysis 2 - Define price and non-price competition. - Identify relevant price and non-price strategies suitable for Cineworld. - Application – competitors such as Odeon and Vue cinema. Benefits and limitations of each can be taken as Knowledge, Application and Analysis, or Evaluation - Price wars – lowering the price to attract customers from competition – increasing sales. - Predatory pricing – where a firm prices below AVC/AC of competitors – enables competitors to be driven out of the market and increases sales enabling the firm to drive up prices later. - Price leadership – where Cineworld sets its price first in order for competitors to follow – leading price may enable the firm to under-cut competitors for a time or push up the price as competitors follow. - Limit pricing – keep price low – to avoid firms contesting/entering market – lower price means higher output and sales, lack of competition means less fierce advertising and price reduction needed. Types of non-price competition - Advertising – this will raise awareness, interest, desire and action to increase sales of cinema tickets. - Branding – investing in the image, logo, slogan of the business – to build trust amongst customers.	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(d) continued	Evaluation 4 - Price wars – competitors may lower the price also meaning customers are the only ones to benefit, profit margins fall. - Predatory pricing – could get into legal trouble for trying to shut down competitor, if they have reserves they could last a long time driving down your own profits/ reserves. - Price leadership – Cineworld may increase price but competitors may not. - Limit pricing – low price means higher output but lower revenue and profit as not profit maximising. Types of non-price competition - Advertising – this will cost money, and may just encourage competitors to do the same making it more costly to remain in market. - Branding – may be a waste of funds as it may not encourage trust or brand loyalty. Other evaluation Game theory - Competitors likely to react and adjust price or advertising budgets/so could end up with both with lower price or higher costs in terms of advertising budget. - Pay off matrix may be used to show worst outcome for all. - Magnitude – depends on amount of price adjustment and advertising budget. - Prioritisation of the significance of each factor – whether pricing and non-price will be most important - Short run little change – people do not adjust to price changes straight away/long run may take time for benefits to emerge, e.g. branding.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 • Competition Commission (CC) concluded merger would lead to higher prices for customers in Aberdeen, Bury St Edmunds and Cambridge. • Diagram, e.g. monopoly or comparing monopoly and perfect competition. • Increased producer surplus/decreased consumer surplus – fairness. • Deadweight loss from monopolist. • Lack of productive efficiency/allocative efficiency ($P > MC$). • X-inefficiency/organisational slack.	(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • Cineworld operates mainly large out-of-town-centre cinemas, Picturehouse's cinemas tend to be smaller cinemas located in city centres – not direct competitors. • Other indirect competitors – bowling etc. • Dynamic efficiency, increased profits may be used to innovate. • National pricing operated where it is the same for all locations. • Management monitored by head office so unlikely to develop X-inefficiency. • May revenue maximise or sales maximise due to divorce of ownership.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Q2.

Question Number	Answer	Mark
(a)	Application 2 Application: 1 mark for applying the formula and 1 mark for correct answer Calculate the PED $5/11$ (1) = 0.4545 (1) NB if the answer given is 0.4545 award 2 marks.	(2)

Question Number	Answer	Mark
(b)	D	(1)

Question Number	Answer	Mark
(c)	Knowledge 1, Analysis 1 Knowledge/understanding: 1 mark for identifying the impact of the change in price and quantity Analysis: 1 mark for linking this to overall impact on total revenue, e.g. - relatively inelastic demand so additional revenue earned from additional price will be greater than lost revenue due to lost quantity (1) so as price rises total revenue rises (1) - may be illustrated diagrammatically – showing total revenue rising as price rises (1) with inelastic demand. (1)	(2)

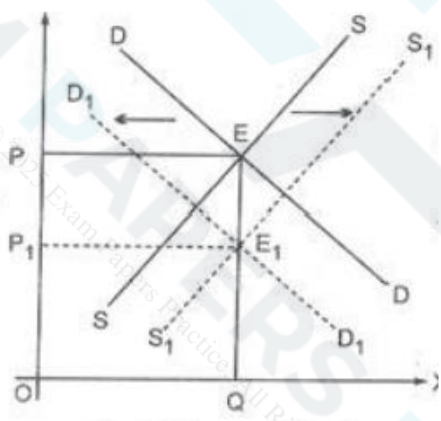
Q3.

Question Number	Answer	Mark
(a)	Application 2 Application: 1 mark for calculating both the percentage change in quantity or the percentage change in earnings. 1 mark for correct answer Percentage change in quantity = $0.32/1.94 \times 100 = 16.5$ (1) And Percentage change in earnings = $16.8/500.7 \times 100 = 3.4$ (1) YED = $16.5/3.4 = 4.9$ (1) NB if the answer given is 4.8 to 5 award 2 marks.	(2)

Question Number	Answer	Mark
(b)	D	(1)

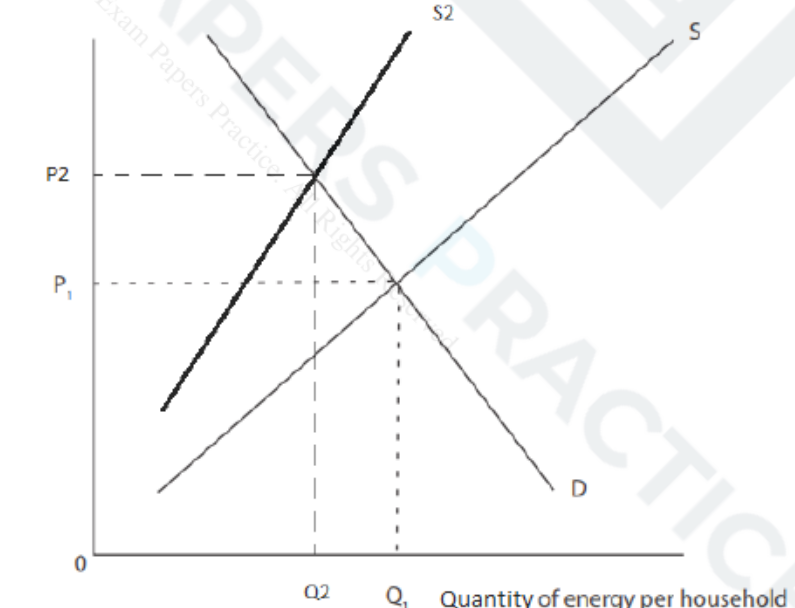
Question Number	Answer	Mark
(c)	Knowledge 1, Analysis 1 Knowledge/understanding: 1 mark for identifying the impact of the increased income on increasing new car sales Analysis: 1 mark for link to slowing growth and how the rate car sales grow will slow, e.g. New car sales will increase in each of the years (1) but the rate at which they increase will slow (1).	(2)

Q4.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2 Knowledge - Supply and demand diagram annotated to show: a right shift in the supply curve (1) and a left shift in the demand curve (1) Application - New lower price (1) - New equilibrium price at D1 and S1 as illustrated below, for example (1) 	(4)
Question Number	Answer	Mark
(b)	Knowledge 1 The only correct answer is C <i>A is not correct because they have mistakenly only calculated the change in price as a percentage change</i> <i>B is not correct because they have mistakenly calculated -the new price divided by the original price initially</i> <i>D is not correct because they have mistakenly calculated the change in price by the new price.</i>	(1)

Q5.

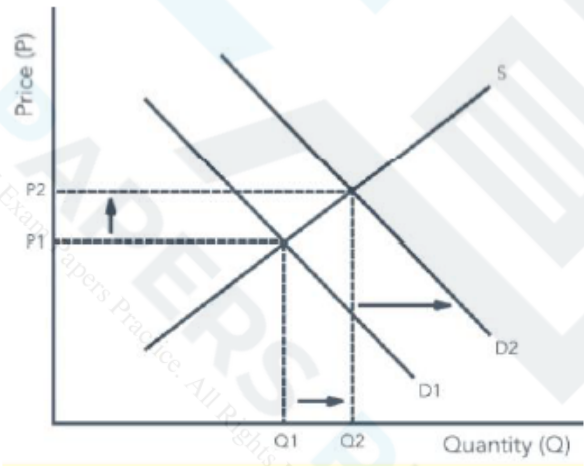
Question Number	Answer	Mark
(a)	Application 1 The only correct answer is A <i>B is not correct because the candidate has mistakenly divided 1070 by 0.5</i> <i>C is not correct because the candidate has mistakenly multiplied 1070 by 1.005</i> <i>D is not correct because the candidate has mistakenly multiplied 1070 by 1.05</i>	(1)

Question Number	Answer	Mark
(b)	Application 2 Average annual household bill (£)  Quantity of energy per household 1 mark for skewed (pivoted) leftward shift of supply curve 1 mark for new equilibrium NB if a parallel leftward shift of supply with correct new equilibrium maximum 1 mark	(2)

Question Number	Answer	Mark
(c)	Knowledge 1 Knowledge 1 mark for definition, e.g. an indirect tax is imposed on producers (suppliers) by the government (1) tax on good or service which is added to their price (1) tax on consumer expenditure (1)	(1)

Q6.

Question Number	Answer	Mark
(a)	Analysis 1 The only correct answer is D <i>A is not correct because fixed costs are not related directly with output in the short-term</i> <i>B is not correct there is no information available on income changes</i> <i>C is not correct because there is no information available on price changes</i>	(1)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2 Knowledge - Correctly labelled original demand and supply (1) annotated to show the demand curve shifted to the right (1) Application As a result of demand shifting right: - Price increase (1) - Output increase (1) - Original and new equilibrium points identified (1+1) ALLOW horizontal supply Award a maximum of three marks if supply also shifts Diagram e.g. 	(4)

Q7.

Question Number	Answer	Mark
(a)	Application 2 Application: 1 mark for likely understanding e.g. • \$40 x 300 seats (1) OR • \$12 000 per show (1) OR • \$12 000 x 5 (1) Answer = \$60 000 Award 2 marks for correct answer (60 000)	(2)

Question Number	Answer	Mark
(b)	Application 2 Application: 1 mark for understanding e.g. - Subsidy is 57.5% (1) OR 100% - 30% (tickets sales) - 12.5% (charitable donations) OR 57.5% x \$ 200 000 (1) OR 0.575 x 200 000 (1) Answer = \$115 000 Award 2 marks for correct answer accept a range (114 000 - 116 000)	(2)

Question Number	Answer	Mark
(c)	Analysis 1 The only correct answer is D <i>A is not correct because this is sales maximisation</i> <i>B is not correct because this is identifying allocative efficiency</i> <i>C is not correct because this is describing perfect competition market structure.</i>	(1)

Q8.

Question Number	Answer	Mark
(a)	Application 2 Application: 1 mark for appropriate calculation e.g. $£2 \times 73 \text{ million} = £146 \text{ million}$ (1) OR $£0.24 \times 73 \text{ million} = £17.52 \text{ million}$ (1) OR $£1.76 \times 73 \text{ million}$ (1) OR correct answer with no millions (1) Answer = £128.48 million (accept between £128 million and £129 million) Award 2 marks for correct answer (128m)	(2)

Question Number	Answer	Mark
(b)	Knowledge 1, Analysis 1 Knowledge: 1 mark for e.g. - The demand for use of gaming machines is likely to increase (1) - It is a substitute for national lottery tickets (1) - Substitutes have a positive cross elasticity of demand (1) Analysis: 1 mark for e.g. - Increase by 128% in response to a 100% increase in the price of national lottery tickets (1) 1.28 cross elasticity of demand indicates they are close substitutes (1)	(2)

Question Number	Answer	Mark
(c)	The only correct answer is C A is not correct because the flood defence scheme is non-exclusive B is not a private good because flood defence scheme is non-exclusive so you are unable to charge private individuals that benefit from the scheme D is not correct because it is a distractor in lacking meaning but also indicating that the candidate is confusing rivalry with the non-rivalrous nature of public goods	(1)

Q9.

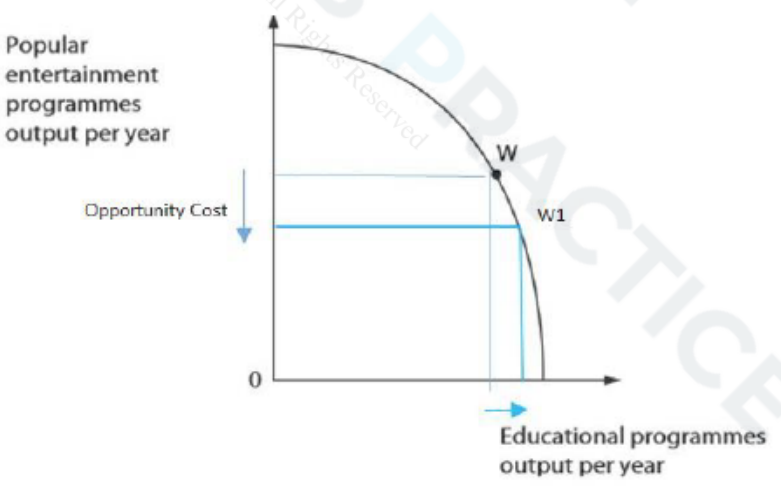
Question Number	Answer	Mark
(a)	Application 2 Application: 1 mark for likely understanding e.g. • $5\% \times -1.37$ (1) OR • $5\% \times 1.37$ (1) Answer = - 6.85% Award 2 marks for correct answer (accept any in the range of 6.8 to 7 or -6.8 to -7)	(2)

Question Number	Answer	Mark
(b)	Application 2 Application: 1 mark for likely understanding e.g. • $5\% \times 0.63$ (1) OR • $5\% \times -0.63$ (1) OR Answer = 3.15% Award 2 marks for correct answer (accept any in the range of 3 to 3.2)	(2)

Question Number	Answer	Mark
(c)	Knowledge 1 The only correct answer is D <i>A is not correct because demand is price-elastic and XED is positive (substitute)</i> <i>B is not correct because demand is price-elastic and XED is less than +1</i> <i>C is not correct because XED is positive (substitute)</i>	(1)

Q10.

Question Number	Answer	Mark
(a)	Knowledge 1 Knowledge/understanding 1 mark for definition, e.g. • Measure of the welfare (utility) that people gain from consuming goods and services • Difference between the total amount that consumers are willing and able to pay and the total amount that they actually pay. • Diagram showing area under the demand curve and above the market equilibrium price.	(1)

Question Number	Answer	Mark
(b)	Application 2 Application 2 marks for, e.g. An additional point to the right of W indicating movement of resources towards educational programmes (1) and the resulting opportunity cost, showing a fall in output of popular entertainment programmes (1). The following diagram earns 2 marks: 	(2)

Question Number	Answer	Mark
(c)	The only correct answer is C <i>A is not correct because this is providing an example of government finances rather than being non-excludable or non-rivalrous as a definition of a public good.</i> <i>B is not correct because this is confusing popularity with public goods, rather than being non-excludable or non-rivalrous as a definition of a public good.</i> <i>D is not correct because this is confusing benefits to third parties as members of the public rather than being non-excludable or non-rivalrous as a definition of a public good.</i>	(1)

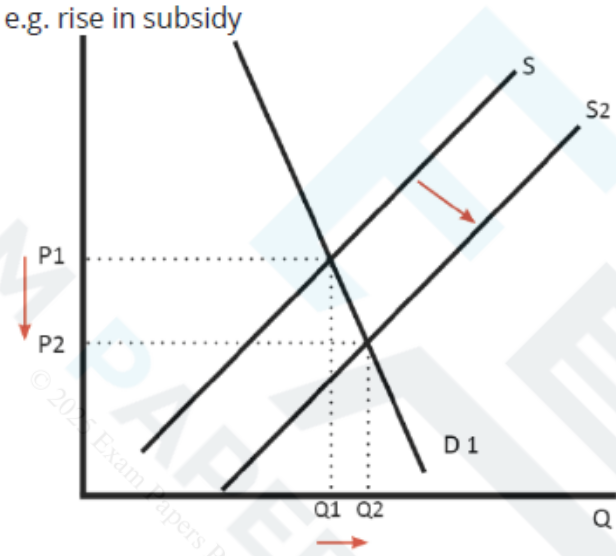
Q11.

Question Number	Answer	Mark
(a)	Application 2 Application: Calculation of original opportunity cost of producing 50 capital goods is $120 - 100 = 20$ consumer goods. (1 mark) Calculation of new opportunity cost of 50 capital goods is $170 - 140 = 30$ consumer goods. (1 mark)	(2)

Question Number	Answer	Mark
(b)	The only correct answer is D <i>A is not correct because this would be describing a change in actual output such as W to V from inside the PPF XY</i> <i>B is not correct because PPF shows potential output and not the cost of output</i> <i>C is not correct because PPF shows potential output and not the demand for a good or service</i>	(1)

Question Number	Answer	Mark
(c)	Knowledge 1, Analysis 1 Knowledge/understanding: 1 mark for e.g. Position W represents an • inefficient allocation of resources (1) • unemployment of resources (1) Analysis: 1 mark for e.g. • It is possible to increase output of both consumer goods and capital goods by using unemployed resources (1) • The economy could produce at either 100 or 140 consumer goods but is currently producing 80 (1) • The country is operating inside the PPF (1)	(2)

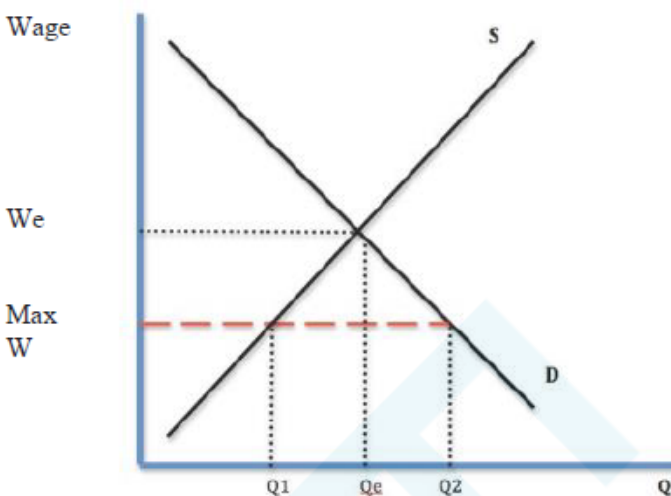
Q12.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 1, Analysis 3 Knowledge and Analysis: (up to 4 marks) Correctly labelled diagram annotated to show a shift right in the supply curve (2) e.g. rise in subsidy  NB: also accept a double shift right indicating an increase in subsidy. NB: also accept a rise in price, given prices annually linked to RPI, but less than it would have been. NB: 2 KAA marks are reserved for an accurate diagram	
	Explanation that subsidy increase is a rise in financial assistance OR cut in cost of production OR additional cash grant (2) Explanation that decreasing costs borne by producers is passed on to consumers as a fall in rail fares OR excess supply results in fall in rail fares (2) Application: 1 for reference to Figure 1 e.g. subsidy rises around £11 million (1) subsidy rises from £277 million to £287.9 million (1) subsidy increased by 3.9% (1)	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Analysis 2, Application 2, Evaluation 2 Knowledge/understanding: 2 marks for identification of two non price determinants (1+1) e.g. • Change in taste and preference/quality of service • Changes in income • Price of substitute has risen/substitute less attractive • Price of complement has fallen/complement more attractive • Increased population/migration • Lower price in real terms • Lower prices – special offers/off-peak/forward booking.	
	Analysis: 1 mark for linked explanation of each identified determinant to demand increasing/shifting right (this may be shown on a diagram) (1+1) Increase in population moved to commuter areas/improvement in rail stock e.g. wifi access. Income elasticity of demand Close substitute e.g. car travel, XED >+1 Close complement e.g. on-board meal, XED >-1 Application: 2 marks for reference to Figure 2 (1+1) or other data (1), e.g. • Approximately 50 billion rail passenger kilometres per year (1) to 65 billion (1) or increased by 15 billion rail passenger kilometres per year (2) or Increased by 30% (2) • Economic recovery post 2008 crisis (1) • Cost of car travel/road congestion (1) • Quality of train meals/on board Wi-Fi access (1)	

	Evaluation: 2 marks for two evaluative comments, OR 2 marks for identification and linked development of one evaluative comment, evaluating significance of factors or another reason as a counter-argument e.g. • Delays and cancellations (1) • Recession 2008- 2010 (1) • Car travel still cheaper/train delays and cancellations (1) • Rail meals expensive/congested trains so can't work on board (1) • Increased home working • Rail fares have risen (1)	(8)
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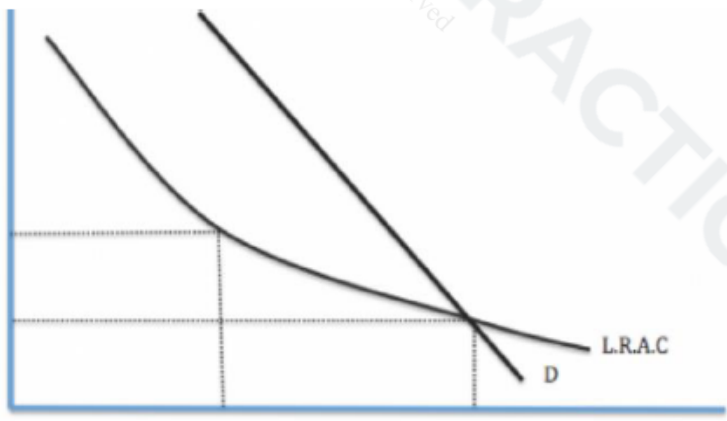
Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 2 Positive effect on protection of employees: • Safety of employees maybe better protected • Trade unions will succeed in securing higher wages in negotiations with government rather than private firms. Rail employees become subject to public sector wage setting. Extension in supply of labour. • Greater job security • Improved integration between state owned rail network and rail operators resulting in rail expansion. • Reduced inequality in pay between employees. • Private sector firms may cut less profitable route whereas government will continue these routes protecting employees Negative effect on protection of employees: • 'Maximum wage on executive managers pay' may be implemented – causing shortages with specialist managers • Nationalisation may not 'provide a wider choice'/quality resulting in lower levels of employment • Lack of funds for investment from the government may reduce the long-term attractiveness of railways causing service to be cut in future	

	NB: Allow relevant diagram e.g.  NB: Negative effects can be seen as KAA and positive effects as evaluation or vice versa. NB: for a Level 3 response there must be reference to the context of nationalisation of the rail industry. NB: for a Level 3 response there must be reference to the effect on employees.	(6)
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3-4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5-6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 - Limited effect since many train operators such as East Coast now in the public sector - Rail usage which increased under privatisation may decline causing suppliers profits and employees wages to decline - Public or private sector ownership may make no difference to executive managers pay - Wages may not increase faster under state ownership compared to private sector. - Level of personnel could be cut substantially under nationalisation if the government has to make spending cuts. - Wages may not increase faster under state ownership compared to private sector. NB: Negative effects can be seen as KAA and positive effects as evaluation or vice versa. NB: for a Level 2 response there must be reference to employees in context.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 2, Application 2, Analysis 4 Case for rail network being a natural monopoly • Define natural monopoly • Application – rail network was renationalised in 2001 after privatisation in 1993 promised competition. Need for integrated national rail network of 32 000km for 2 500 stations, requires enormous investment. • Large infrastructure cost means adding one more passenger or one more train means AC continues to fall. • Little additional costs associated with an additional customer or train running. • Operating one rail track minimises average costs, rail network's LRAC continues to fall as high fixed costs associated with laying track and building new stations. • Allocative efficiency is met at close to large share of market demand. Minimum efficiency scale will require large subsidies to fund off peak services and cover losses. • Average cost would be very high for having two or more rail companies with wasteful duplication of multiple tracks – resulting in higher rail fares. • Allow relevant diagram e.g.	
	AC  Output • Only one rail network company can fully exploit large internal economies of scale – technical, commercial, managerial. NB: Case for can be seen as KAA and positive case against as evaluation or vice versa.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3-5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6-8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(d) continued	Evaluation 4 Case for rail network not being a natural monopoly • Government support for competition: “companies can bid to build new rail lines to upgrade the railway.” • Train operating companies already compete to run trains on lines • Rail lines and stations could be leased out to competing firms • Potential diseconomies of scale in rail infrastructure – managerial slack, closure of rail lines. • Technological advances may result in competing rail lines. NB: Case for can be seen as KAA and positive case against as evaluation or vice versa.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 Price discrimination – charging different prices to different groups of consumers for the same service e.g. 16-25 railcard users vs adult fare users. Allow peak and off-peak as service from A to B is the same, price discrimination just provides an incentive to travel slightly later. Lower rail fares for 16-25 year olds Young persons' may increase their consumer surplus as a result – prices may move towards being more allocatively efficient $P=MC$. May find bargain fares - £35.80 cheaper as in Figure 3. Allows lower income groups to consume service – seek employment. More choice for rail travellers: Price discrimination enables TOCs to make more revenue so they can cross subsidise and provide discount fares for seats that would not have been taken. In some cases without price discrimination rail operators may go out of business – preventing rail travel.	

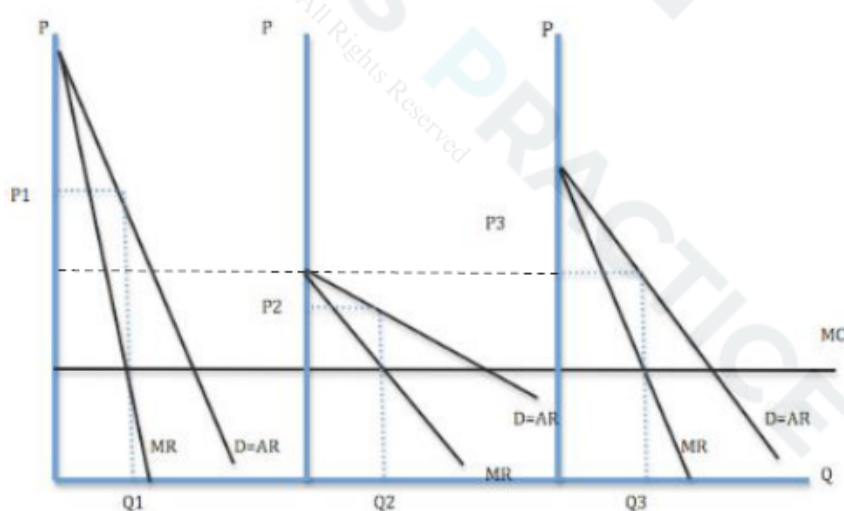
- **Avoids overcrowding**

Trail operators can use price discrimination to manage demand, preventing overcrowded train in peak travel times and encouraging rail travel by 16-25 year olds in less popular times off peak.

- **Improved customer service**

Rail travellers receive a better-quality travelling experience thanks to additional revenue raised through price discrimination. Investment in new trains, station services, online ticketing, customer information, Improved food and drink service, Wi-Fi access, seating and storage space.

- **Price discrimination diagram** showing, for example: Market profit maximising price and output P3 and Q3. At which no service would be provided for 16-25 year old customers. Price discrimination enables a lower price of P2 (£69.50) for young persons' – so consumer surplus increases for consumers with relatively more price elastic demand



NB: Other correct diagrams accepted.

NB: For Level 3 a valid diagram such as above is required. Analysis must link to benefits to consumers. Allow other types of rail fare price discrimination e.g advance booking, Senior Railcard, 16-17 year olds half price travel.

(9)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4-6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7-9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 • Rail operators are price regulated so not pure price makers • Cuts in number of guards may mean adults travel illegally. • Higher rail fares for adults £105.30 in Figure 3 (£35.80 more than 16-25 year olds) resulting in fall in consumer surplus and less allocatively efficient pricing. Overall consumer surplus may be lower. • Despite price discrimination consumers are suffering train cancellations and government is having to rescue rail operators making large losses. • Overcrowding happens on off peak trains – so government/rail operators have no incentive to continue offering young persons' discount. • Deteriorating customer service despite price discrimination. Rail operators using additional revenue to pay out dividends or executive pay rises. • Are rail operators/government aware of PEDs for different groups of consumers.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	5-6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Q13.

Question Number	Answer	Mark
(a)	Application 2 Application $PED = \% \Delta Q_d / \% \Delta P$ price rise = $5/60 \times 100 = 8.3\%$ (1) or $-5.4\%/+8.3\%$ (1) $PED = (-)0.65$ Award 2 marks for $(-)0.65$ NB if answer given is between $(-)0.7$ and $(-)0.6$ award 2 marks.	(2)

Question Number	Answer	Mark
(b)	The only correct answer is A <i>B is not correct because 'population' is a non price determinant of demand not PED determinant</i> <i>C is not correct because 'cost' is a non price determinant of supply not PED determinant</i> <i>D is not correct because 'expected rise in the price' is a non price determinant of demand not PED determinant</i>	(1)

Question Number	Answer	Mark
(c)	Knowledge 1 Knowledge/understanding 1 mark for definition: 'all other things being equal' (1) OR All other factors remain unchanged (1)	(1)

Q14.

Question Number	Answer	Mark
(a)	The only correct answer is D <i>A is not correct because the concept explored is PED not YED</i> <i>B is not correct because as unemployment falls fewer people will use buses</i> <i>C is not correct because XED is concerned with the relationship between complements or substitutes</i>	(1)

Question Number	Answer	Mark
(b)	Knowledge 1, Application 1 and Analysis 1 Knowledge/understanding 1 mark for reason Possible answers include: - Recognition that the lower the PED the less price sensitive the product is (1) quantity demanded does not change significantly with price (1) Application - If, for example, a product with a PED of -0.1 had a 10% price rise, sales volume would fall by 1% (ceteris paribus) (1); so the 10% higher price per unit would far outweigh the 1% reduction in units sold (1) Analysis - The lower a product's price elasticity, the easier it is to boost revenue by increasing the price (1) if the firm raises its price it will increase the value of its sales (1)	(3)

Q15.

Question Number	Answer	Mark
(a)	Application 1 The only correct answer is C <i>A is not correct because they may multiplied 8.75 by -0.1</i> <i>B is not correct because they have mistakenly taken 2017 as the original year in the percentage change calculation</i> <i>D is not correct because the percentage change in quantity demanded has not been multiplied by -0.1.</i>	(1)

Question Number	Answer	Mark
(b)	Knowledge 1, Application 1, Analysis 1 Knowledge/Understanding 1 mark for recognition of the effect of a change in income on the demand Application 1 mark for application to Freddo, e.g. • YED is negative (1) • Freddos is an inferior good (1) • YED is inelastic Analysis 1 mark for linked development, e.g. • Sales revenues fall as incomes rise or • Sales revenue rise when incomes fall • Company should plan to produce less when the economy does well or • Company should plan to produce more when economy/incomes are not growing	(3)